# Propositional and First-Order Logic

Chapter 7.4-7.5, 7.7, 8.1–8.3, 8.5 (Russel and Norvig textbook)

Some material adopted from notes and slides by Tim Finin, Marie desJardins, Andreas Geyer-Schulz and Chuck Dyer

# Propositional Logic

# Propositional logic

- **Logical constants**: true, false
- **Propositional symbols**: P, Q, S, ... (**atomic sentences**)
- Wrapping **parentheses**: ( … )
- Sentences are combined by **connectives**:

| | | |
|---|---|---|
| $\wedge$ | ...and | [conjunction] |
| $\vee$ | ...or | [disjunction] |
| $\Rightarrow$ | ...implies | [implication / conditional] |
| $\Leftrightarrow$ | ..is equivalent | [biconditional] |
| $\neg$ | ...not | [negation] |

- **Literal**: atomic sentence or negated atomic sentence

# Examples of PL sentences

- P means “It is hot.”
- Q means “It is humid.”
- R means “It is raining.”
- $(P \wedge Q) \rightarrow R$

  “If it is hot and humid, then it is raining”
- $Q \rightarrow P$

  “If it is humid, then it is hot”
- A better way:

  Hot = “It is hot”

  Humid = “It is humid”

  Raining = “It is raining”

# Propositional logic (PL)

- A simple language useful for showing key ideas and definitions
- User defines a set of propositional symbols, like P and Q.
- User defines the **semantics** of each propositional symbol:
  - P means “It is hot”
  - Q means “It is humid”
  - R means “It is raining”
- A sentence (well formed formula) is defined as follows:
  - A symbol is a sentence
  - If S is a sentence, then $\neg S$ is a sentence
  - If S is a sentence, then (S) is a sentence
  - If S and T are sentences, then $(S \vee T)$, $(S \wedge T)$, $(S \rightarrow T)$, and $(S \leftrightarrow T)$ are sentences
  - A sentence results from a finite number of applications of the above rules

# Some terms

- The meaning or **semantics** of a sentence determines its **interpretation**.
- Given the truth values of all symbols in a sentence, it can be "evaluated" to determine its **truth value** (True or False).
- A **model** for a KB is a "possible world" (assignment of truth values to propositional symbols) in which each sentence in the KB is True.

# More terms

- A **valid sentence** or **tautology** is a sentence that is True under all interpretations, no matter what the world is actually like or how the semantics are defined. Example: “It’s raining or it’s not raining.”
- An **inconsistent sentence** or **contradiction** is a sentence that is False under all interpretations. The world is never like what it describes, as in “It’s raining and it’s not raining.”
- **P entails Q**, written P |= Q, means that whenever P is True, so is Q. In other words, all models of P are also models of Q.

# Truth tables

*And*

| $p$ | $q$ | $p \cdot q$ |
|---|---|---|
| $T$ | $T$ | $T$ |
| $T$ | $F$ | $F$ |
| $F$ | $T$ | $F$ |
| $F$ | $F$ | $F$ |

*Or*

| $p$ | $q$ | $p \vee q$ |
|---|---|---|
| $T$ | $T$ | $T$ |
| $T$ | $F$ | $T$ |
| $F$ | $T$ | $T$ |
| $F$ | $F$ | $F$ |

*If . . . then*

| $p$ | $q$ | $p \rightarrow q$ |
|---|---|---|
| $T$ | $T$ | $T$ |
| $T$ | $F$ | $F$ |
| $F$ | $T$ | $T$ |
| $F$ | $F$ | $T$ |

*Not*

| $p$ | $\sim p$ |
|---|---|
| $T$ | $F$ |
| $F$ | $T$ |

# Truth tables II

The five logical connectives:

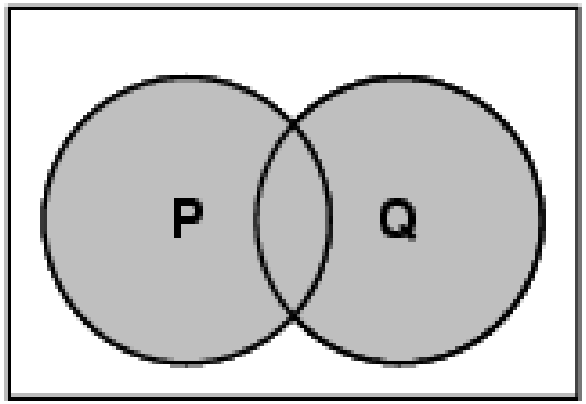
| $P$ | $Q$ | $\neg P$ | $P \wedge Q$ | $P \vee Q$ | $P \Rightarrow Q$ | $P \Leftrightarrow Q$ |
|---|---|---|---|---|---|---|
| *False* | *False* | *True* | *False* | *False* | *True* | *True* |
| *False* | *True* | *True* | *False* | *True* | *True* | *False* |
| *True* | *False* | *False* | *False* | *True* | *False* | *False* |
| *True* | *True* | *False* | *True* | *True* | *True* | *True* |

A complex sentence:

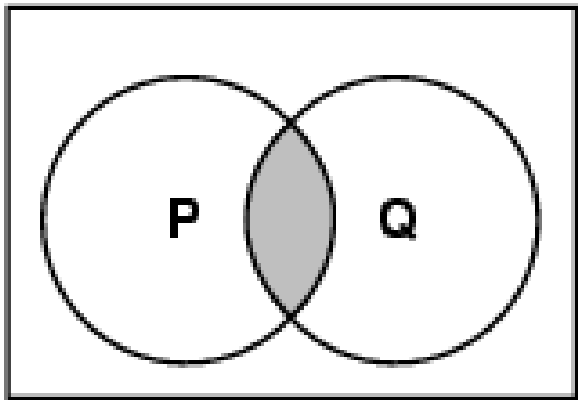
| $P$ | $H$ | $P \vee H$ | $(P \vee H) \wedge \neg H$ | $((P \vee H) \wedge \neg H) \Rightarrow P$ |
|---|---|---|---|---|
| *False* | *False* | *False* | *False* | *True* |
| *False* | *True* | *True* | *False* | *True* |
| *True* | *False* | *True* | *True* | *True* |
| *True* | *True* | *True* | *False* | *True* |

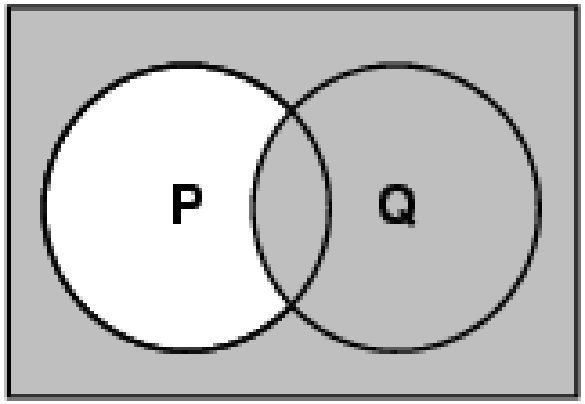
# Models of complex sentences

$P \vee Q$

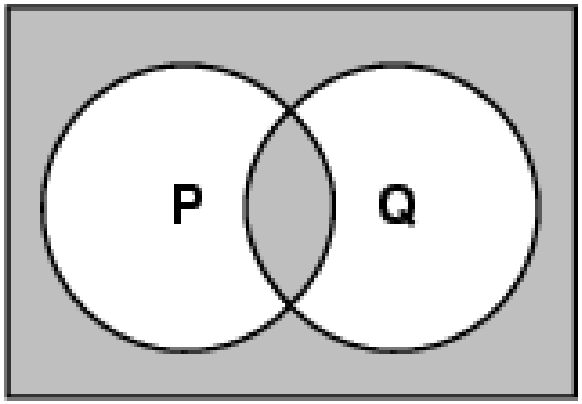



$P \wedge Q$




$P \Rightarrow Q$




$P \Leftrightarrow Q$

# Inference rules

- **Logical inference** is used to create new sentences that logically follow from a given set of predicate calculus sentences (KB).
- An inference rule is **sound** if every sentence X produced by an inference rule operating on a KB logically follows from the KB. (That is, the inference rule does not create any contradictions)
- An inference rule is **complete** if it is able to produce every expression that logically follows from (is entailed by) the KB. (Note the analogy to complete search algorithms.)

# Sound rules of inference

- Here are some examples of sound rules of inference
  - *A rule is sound if its conclusion is true whenever the premise is true*
- Each can be shown to be sound using a truth table

| RULE | PREMISE | CONCLUSION |
|---|---|---|
| Modus Ponens | $A, A \rightarrow B$ | $B$ |
| And Introduction | $A, B$ | $A \wedge B$ |
| And Elimination | $A \wedge B$ | $A$ |
| Double Negation | $\neg\neg A$ | $A$ |
| Unit Resolution | $A \vee B, \neg B$ | $A$ |
| **Resolution** | $A \vee B, \neg B \vee C$ | $A \vee C$ |

# Soundness of modus ponens

| **A** | **B** | **A → B** | **OK?** |
|---|---|---|---|
| True | True | True | √ |
| True | False | False | √ |
| False | True | True | √ |
| False | False | True | √ |

# Soundness of the resolution inference rule

| $\alpha$ | $\beta$ | $\gamma$ | $\alpha \vee \beta$ | $\neg\beta \vee \gamma$ | $\alpha \vee \gamma$ |
|---|---|---|---|---|---|
| *False* | *False* | *False* | *False* | *True* | *False* |
| *False* | *False* | *True* | *False* | *True* | *True* |
| *False* | *True* | *False* | *True* | *False* | *False* |
| *False* | *True* | *True* | *True* | *True* | *True* |
| *True* | *False* | *False* | *True* | *True* | *True* |
| *True* | *False* | *True* | *True* | *True* | *True* |
| *True* | *True* | *False* | *True* | *False* | *True* |
| *True* | *True* | *True* | *True* | *True* | *True* |

# Proving things

- A **proof** is a sequence of sentences, where each sentence is either a premise or a sentence derived from earlier sentences in the proof by one of the rules of inference.
- The last sentence is the **theorem** (also called goal or query) that we want to prove.
- Example for the "weather problem" given above.

| | | |
|---|---|---|
| 1 Humid | Premise | "It is humid" |
| 2 Humid→Hot | Premise | "If it is humid, it is hot" |
| 3 Hot | Modus Ponens(1,2) | "It is hot" |
| 4 (Hot^Humid)→Rain | Premise | "If it's hot & humid, it's raining" |
| 5 Hot^Humid | And Introduction(1,2) | "It is hot and humid" |
| 6 Rain | Modus Ponens(4,5) | "It is raining" |

# Horn sentences

- A **Horn sentence** or **Horn clause** has the form:

  $P1 \wedge P2 \wedge P3 \ldots \wedge Pn \rightarrow Q$

or alternatively

$(P \rightarrow Q) = (\neg P \vee Q)$

  $\neg P1 \vee \neg P2 \vee \neg P3 \ldots \vee \neg Pn \vee Q$

where Ps and Q are non-negated atoms

- To get a proof for Horn sentences, apply Modus Ponens repeatedly until nothing can be done
- We will use the Horn clause form later

# Entailment and derivation

- **Entailment: KB |= Q**
  - Q is entailed by KB (a set of premises or assumptions) if and only if there is no logically possible world in which Q is false while all the premises in KB are true.
  - Or, stated positively, Q is entailed by KB if and only if the conclusion is true in every logically possible world in which all the premises in KB are true.
- **Derivation: KB |- Q**
  - We can derive Q from KB if there is a proof consisting of a sequence of valid inference steps starting from the premises in KB and resulting in Q

# Two important properties for inference

## Soundness: If KB |- Q then KB |= Q

- If Q is derived from a set of sentences KB using a given set of rules of inference, then Q is entailed by KB.
- Hence, inference produces only real entailments, or any sentence that follows deductively from the premises is valid.

## Completeness: If KB |= Q then KB |- Q

- If Q is entailed by a set of sentences KB, then Q can be derived from KB using the rules of inference.
- Hence, inference produces all entailments, or all valid sentences can be proved from the premises.

# Propositional logic is a weak language

- Hard to identify “individuals” (e.g., Mary, 3)
- Can’t directly talk about properties of individuals or relations between individuals (e.g., “Bill is tall”)
- Generalizations, patterns, regularities can’t easily be represented (e.g., “all triangles have 3 sides”)
- First-Order Logic (abbreviated FOL or FOPC) is expressive enough to concisely represent this kind of information

  FOL adds relations, variables, and quantifiers, e.g.,

  - *“Every elephant is gray”:* $\forall$ x (elephant(x) $\rightarrow$ gray(x))
  - *“There is a white alligator”:* $\exists$ x (alligator(X) ^ white(X))

# Example

- Consider the problem of representing the following information:
  - Every person is mortal.
  - Confucius is a person.
  - Confucius is mortal.
- How can these sentences be represented so that we can infer the third sentence from the first two?

# Example II

- In PL we have to create propositional symbols to stand for all or part of each sentence. For example, we might have:

  P = “person”; Q = “mortal”; R = “Confucius”

- so the above 3 sentences are represented as:

  $P \rightarrow Q$; $R \rightarrow P$; $R \rightarrow Q$

- Although the third sentence is entailed by the first two, we needed an explicit symbol, R, to represent an individual, Confucius, who is a member of the classes “person” and “mortal”
- To represent other individuals we must introduce separate symbols for each one, with some way to represent the fact that all individuals who are “people” are also “mortal”

# The “Hunt the Wumpus” agent

- Some atomic propositions:

  S12 = There is a stench in cell (1,2)

  B34 = There is a breeze in cell (3,4)

  W22 = The Wumpus is in cell (2,2)

  V11 = We have visited cell (1,1)

  OK11 = Cell (1,1) is safe.

  etc

- Some rules:

  (R1) $\neg S11 \rightarrow \neg W11 \wedge \neg W12 \wedge \neg W21$

  (R2) $\neg S21 \rightarrow \neg W11 \wedge \neg W21 \wedge \neg W22 \wedge \neg W31$

  (R3) $\neg S12 \rightarrow \neg W11 \wedge \neg W12 \wedge \neg W22 \wedge \neg W13$

  (R4) $S12 \rightarrow W13 \vee W12 \vee W22 \vee W11$

  etc

- Note that the lack of variables requires us to give similar rules for each cell

# After the third move

- We can prove that the Wumpus is in (1,3) using the four rules given.
- See R&N section 7.5

| | | | |
|---|---|---|---|
| 1,4 | 2,4 | 3,4 | 4,4 |
| 1,3<br>W! | 2,3 | 3,3 | 4,3 |
| 1,2<br>A<br>S<br>OK | 2,2<br>OK | 3,2 | 4,2 |
| 1,1<br>V<br>OK | 2,1<br>B<br>V<br>OK | 3,1<br>P! | 4,1 |

A = *Agent*
B = *Breeze*
G = *Glitter, Gold*
OK = *Safe square*
P = *Pit*
S = *Stench*
V = *Visited*
W = *Wumpus*

# Proving W13

- Apply MP with ¬S11 and R1:
  ¬ W11 ^ ¬ W12 ^ ¬ W21
- Apply And-Elimination to this, yielding 3 sentences:
  ¬ W11, ¬ W12, ¬ W21
- Apply MP to ~S21 and R2, then apply And-elimination:
  ¬ W22, ¬ W21, ¬ W31
- Apply MP to S12 and R4 to obtain:
  W13 ˇ W12 ˇ W22 ˇ W11
- Apply Unit resolution on (W13 ˇ W12 ˇ W22 ˇ W11) and ¬W11:
  W13 ˇ W12 ˇ W22
- Apply Unit Resolution with (W13 ˇ W12 ˇ W22) and ¬W22:
  W13 ˇ W12
- Apply UR with (W13 ˇ W12) and ¬W12:
  W13
- QED

# Problems with the propositional Wumpus hunter

- Lack of variables prevents stating more general rules
  - We need a set of similar rules for each cell
- Change of the KB over time is difficult to represent
  - Standard technique is to index facts with the time when they're true
  - This means we have a separate KB for every time point

# Summary

- The process of deriving new sentences from old one is called **inference**.
  - **Sound** inference processes derives true conclusions given true premises
  - **Complete** inference processes derive all true conclusions from a set of premises
- A **valid sentence** is true in all worlds under all interpretations
- If an implication sentence can be shown to be valid, then—given its premise—its consequent can be derived
- Different logics make different **commitments** about what the world is made of and what kind of beliefs we can have regarding the facts
  - Logics are useful for the commitments they do not make because lack of commitment gives the knowledge base engineer more freedom
- **Propositional logic** commits only to the existence of facts that may or may not be the case in the world being represented
  - It has a simple syntax and simple semantics. It suffices to illustrate the process of inference
  - Propositional logic quickly becomes impractical, even for very small worlds

# First-Order Logic

# Outline

- First-order logic
  - Properties, relations, functions, quantifiers, …
  - Terms, sentences, axioms, theories, proofs, …
- Extensions to first-order logic
- Logical agents
  - Reflex agents
  - Representing change: situation calculus, frame problem
  - Preferences on actions
  - Goal-based agents

# First-order logic

- First-order logic (FOL) models the world in terms of
  - **Objects,** which are things with individual identities
  - **Properties** of objects that distinguish them from other objects
  - **Relations** that hold among sets of objects
  - **Functions,** which are a subset of relations where there is only one "value" for any given "input"
- Examples:
  - Objects: Students, lectures, companies, cars ...
  - Relations: Brother-of, bigger-than, outside, part-of, has-color, occurs-after, owns, visits, precedes, ...
  - Properties: blue, oval, even, large, ...
  - Functions: father-of, best-friend, second-half, one-more-than ...

# User provides

- **Constant symbols,** which represent individuals in the world
  - Mary
  - 3
  - Green
- **Function symbols,** which map individuals to individuals
  - father-of(Mary) = John
  - color-of(Sky) = Blue
- **Predicate symbols,** which map individuals to truth values
  - greater(5,3)
  - green(Grass)
  - color(Grass, Green)

# FOL Provides

- **Variable symbols**
  - E.g., x, y, foo
- **Connectives**
  - Same as in PL: not (¬), and (^), or (˅), implies (→), if and only if (biconditional ↔)
- **Quantifiers**
  - Universal **∀x** or **(Ax)**
  - Existential **∃x** or **(Ex)**

# Sentences are built from terms and atoms

- A **term** (denoting a real-world individual) is a constant symbol, a variable symbol, or an n-place function of n terms.

  x and $f(x_1, ..., x_n)$ are terms, where each $x_i$ is a term.

  A term with no variables is a **ground term**
- An **atomic sentence** (which has value true or false) is an n-place predicate of n terms
- A **complex sentence** is formed from atomic sentences connected by the logical connectives:

  $\neg P$, $P \vee Q$, $P \wedge Q$, $P \rightarrow Q$, $P \leftrightarrow Q$ where P and Q are sentences
- A **quantified sentence** adds quantifiers $\forall$ and $\exists$
- A **well-formed formula (wff)** is a sentence containing no "free" variables. That is, all variables are "bound" by universal or existential quantifiers.

  $(\forall x)P(x,y)$ has x bound as a universally quantified variable, but y is free.

# Quantifiers

- **Universal quantification**
  - $(\forall x)P(x)$ means that P holds for **all** values of x in the domain associated with that variable
  - E.g., $(\forall x)$ dolphin(x) $\rightarrow$ mammal(x)
- **Existential quantification**
  - $(\exists x)P(x)$ means that P holds for **some** value of x in the domain associated with that variable
  - E.g., $(\exists x)$ mammal(x) $\wedge$ lays-eggs(x)
  - Permits one to make a statement about some object without naming it

# Quantifiers

- Universal quantifiers are often used with “implies” to form “rules”:
  - $(\forall x)$ student(x) $\rightarrow$ smart(x) means “All students are smart”
- Universal quantification is *rarely* used to make blanket statements about every individual in the world:
  - $(\forall x)$student(x)$\wedge$smart(x) means “Everyone in the world is a student and is smart”
- Existential quantifiers are usually used with “and” to specify a list of properties about an individual:
  - $(\exists x)$ student(x) $\wedge$ smart(x) means “There is a student who is smart”
- A common mistake is to represent this English sentence as the FOL sentence:
  - $(\exists x)$ student(x) $\rightarrow$ smart(x)
  - But what happens when there is a person who is *not* a student?

# Quantifier Scope

- Switching the order of universal quantifiers *does not* change the meaning:
  - $(\forall x)(\forall y)P(x,y) \leftrightarrow (\forall y)(\forall x)\, P(x,y)$
- Similarly, you can switch the order of existential quantifiers:
  - $(\exists x)(\exists y)P(x,y) \leftrightarrow (\exists y)(\exists x)\, P(x,y)$
- Switching the order of universals and existentials *does* change meaning:
  - Everyone likes someone: $(\forall x)(\exists y)\, likes(x,y)$
  - Someone is liked by everyone: $(\exists y)(\forall x)\, likes(x,y)$

# Connections between All and Exists

We can relate sentences involving $\forall$ and $\exists$ using De Morgan's laws:

$$(\forall x)\ \neg P(x) \leftrightarrow \neg(\exists x)\ P(x)$$

$$\neg(\forall x)\ P \leftrightarrow (\exists x)\ \neg P(x)$$

$$(\forall x)\ P(x) \leftrightarrow \neg\ (\exists x)\ \neg P(x)$$

$$(\exists x)\ P(x) \leftrightarrow \neg(\forall x)\ \neg P(x)$$

# Quantified inference rules

- Universal instantiation
  - $\forall x\ P(x) \therefore P(A)$
- Universal generalization
  - $P(A) \wedge P(B) \ldots \therefore \forall x\ P(x)$
- Existential instantiation
  - $\exists x\ P(x) \therefore P(F)$ ← **skolem constant F**
- Existential generalization
  - $P(A) \therefore \exists x\ P(x)$

# Universal instantiation (a.k.a. universal elimination)

- If $(\forall x)\ P(x)$ is true, then $P(C)$ is true, where C is *any* constant in the domain of x
- Example:

  $(\forall x)$ eats(Ziggy, x) $\Rightarrow$ eats(Ziggy, IceCream)
- The variable symbol can be replaced by any ground term, i.e., any constant symbol or function symbol applied to ground terms only

# Existential instantiation (a.k.a. existential elimination)

- From $(\exists x)\ P(x)$ infer $P(c)$
- Example:
  - $(\exists x)$ eats(Ziggy, x) $\rightarrow$ eats(Ziggy, Stuff)
- Note that the variable is replaced by a **brand-new constant** not occurring in this or any other sentence in the KB
- Also known as skolemization; constant is a **skolem constant**
- In other words, we don't want to accidentally draw other inferences about it by introducing the constant
- Convenient to use this to reason about the unknown object, rather than constantly manipulating the existential quantifier

# Existential generalization (a.k.a. existential introduction)

- If P(c) is true, then (∃x) P(x) is inferred.
- Example

  eats(Ziggy, IceCream) ⇒ (∃x) eats(Ziggy, x)
- All instances of the given constant symbol are replaced by the new variable symbol
- Note that the variable symbol cannot already exist anywhere in the expression

# Translating English to FOL

**Every gardener likes the sun.**

$\forall x\ gardener(x) \rightarrow likes(x,Sun)$

**You can fool some of the people all of the time.**

$\exists x\ \forall t\ \ person(x) \wedge time(t) \rightarrow can\text{-}fool(x,t)$

**You can fool all of the people some of the time.**

$\forall x\ \exists t\ (person(x) \rightarrow time(t) \wedge can\text{-}fool(x,t))$

$\forall x\ (person(x) \rightarrow \exists t\ (time(t) \wedge can\text{-}fool(x,t)))$

Equivalent

**All purple mushrooms are poisonous**.

$\forall x\ (mushroom(x) \wedge purple(x)) \rightarrow poisonous(x)$

**No purple mushroom is poisonous.**

$\neg\exists x\ purple(x) \wedge mushroom(x) \wedge poisonous(x)$

$\forall x\ \ (mushroom(x) \wedge purple(x)) \rightarrow \neg poisonous(x)$

Equivalent

**There are exactly two purple mushrooms**.

$\exists x\ \exists y\ mushroom(x) \wedge purple(x) \wedge mushroom(y) \wedge purple(y) \wedge \neg(x=y) \wedge\ \forall z$
$(mushroom(z) \wedge purple(z)) \rightarrow ((x=z) \vee (y=z))$

**Clinton is not tall.**

$\neg tall(Clinton)$

**X is above Y iff X is on directly on top of Y or there is a pile of one or more other objects directly on top of one another starting with X and ending with Y.**

$\forall x\ \forall y\ above(x,y) \leftrightarrow (on(x,y) \vee\ \exists z\ (on(x,z) \wedge above(z,y)))$

# Monty Python and The Art of Fallacy

Cast

- **Sir Bedevere the Wise**, master of (odd) logic
- **King Arthur**
- **Villager 1**, witch-hunter
- **Villager 2**, ex-newt
- **Villager 3**, one-line wonder
- **All**, the rest of you scoundrels, mongrels, and nere-do-wells.

# An example from Monty Python by way of Russell & Norvig

- **FIRST VILLAGER:** We have found a witch. May we burn her?
- **ALL:** A witch! Burn her!
- **BEDEVERE:** Why do you think she is a witch?
- **SECOND VILLAGER:** She turned *me* into a newt.
- **B:** A newt?
- **V2** *(after looking at himself for some time)*: I got better.
- **ALL:** Burn her anyway.
- **B:** Quiet! Quiet! There are ways of telling whether she is a witch.

# Monty Python cont.

- **B:** Tell me… what do you do with witches?
- **ALL:** Burn them!
- **B:** And what do you burn, apart from witches?
- **Third Villager:** …wood?
- **B:** So **why do witches burn**?
- **V2** *(after a beat)*: **because they're made of wood?**
- **B:** Good.
- **ALL:** I see. Yes, of course.

# Monty Python cont.

- **B: So how can we tell if she is made of wood?**
- **V1: Make a bridge out of her.**
- B: Ah… but can you not also make bridges out of stone?
- **ALL:** Yes, of course… um… er…
- **B:** Does wood sink in water?
- **ALL:** No, no, it floats. Throw her in the pond.
- **B:** Wait. Wait… tell me, what also floats on water?
- **ALL:** Bread? No, no no. Apples… gravy… very small rocks…
- **B:** No, no, no,

# Monty Python cont.

- **KING ARTHUR:** A duck!
- *(They all turn and look at Arthur. Bedevere looks up, very impressed.)*
- **B:** Exactly. So… logically…
- **V1** *(beginning to pick up the thread)*: **If she… weighs the same as a duck… she's made of wood**.
- **B:** And therefore?
- **ALL: A witch!**

# Monty Python Fallacy #1

$\forall$ $\forall x$ witch(x) $\rightarrow$ burns(x)

$\forall$ $\forall x$ wood(x) $\rightarrow$ burns(x)

- ---

$\forall$ $\therefore$ $\forall z$ witch(x) $\rightarrow$ wood(x)

- $p \rightarrow q$
- $r \rightarrow q$
- ---
- $p \rightarrow r$

Fallacy: Affirming the conclusion

# Monty Python Near-Fallacy #2

- wood(x) → can-build-bridge(x)
- ----------

∀∴ can-build-bridge(x) → wood(x)

- B: Ah… but can you not also make bridges out of stone?

# Monty Python Fallacy #3

∀ $\forall x$ wood(x) → floats(x)

∀ $\forall x$ duck-weight (x) → floats(x)

- ---

∀ ∴ $\forall x$ duck-weight(x) → wood(x)

- p → q
- r → q
- ---

∀ ∴ r → p

# Monty Python Fallacy #4

$\forall$ $\forall z$ light(z) $\rightarrow$ wood(z)

- light(W)
- ----------

$\forall$ $\therefore$ wood(W) ok……..

- witch(W) $\rightarrow$ wood(W) applying universal instan. to fallacious conclusion #1
- wood(W)
- ----------

$\forall$ $\therefore$ witch(z)

# Example: A simple genealogy KB by FOL

- **Build a small genealogy knowledge base using FOL that**
  - contains facts of immediate family relations (spouses, parents, etc.)
  - contains definitions of more complex relations (ancestors, relatives)
  - is able to answer queries about relationships between people
- **Predicates:**
  - parent(x, y), child(x, y), father(x, y), daughter(x, y), etc.
  - spouse(x, y), husband(x, y), wife(x,y)
  - ancestor(x, y), descendant(x, y)
  - male(x), female(y)
  - relative(x, y)
- **Facts:**
  - husband(Joe, Mary), son(Fred, Joe)
  - spouse(John, Nancy), male(John), son(Mark, Nancy)
  - father(Jack, Nancy), daughter(Linda, Jack)
  - daughter(Liz, Linda)
  - etc.

- **Rules for genealogical relations**
  - $(\forall x,y)$ parent(x, y) $\leftrightarrow$ child (y, x)
    $(\forall x,y)$ father(x, y) $\leftrightarrow$ parent(x, y) $\wedge$ male(x) (similarly for mother(x, y))
    $(\forall x,y)$ daughter(x, y) $\leftrightarrow$ child(x, y) $\wedge$ female(x) (similarly for son(x, y))
  - $(\forall x,y)$ husband(x, y) $\leftrightarrow$ spouse(x, y) $\wedge$ male(x) (similarly for wife(x, y))
    $(\forall x,y)$ spouse(x, y) $\leftrightarrow$ spouse(y, x) (**spouse relation is symmetric**)
  - $(\forall x,y)$ parent(x, y) $\rightarrow$ ancestor(x, y)
    $(\forall x,y)(\exists z)$ parent(x, z) $\wedge$ ancestor(z, y) $\rightarrow$ ancestor(x, y)
  - $(\forall x,y)$ descendant(x, y) $\leftrightarrow$ ancestor(y, x)
  - $(\forall x,y)(\exists z)$ ancestor(z, x) $\wedge$ ancestor(z, y) $\rightarrow$ relative(x, y)
    (related by common ancestry)
    $(\forall x,y)$ spouse(x, y) $\rightarrow$ relative(x, y) (related by marriage)
    $(\forall x,y)(\exists z)$ relative(z, x) $\wedge$ relative(z, y) $\rightarrow$ relative(x, y) (**transitive**)
    $(\forall x,y)$ relative(x, y) $\leftrightarrow$ relative(y, x) (**symmetric**)
- **Queries**
  - ancestor(Jack, Fred) /* the answer is yes */
  - relative(Liz, Joe) /* the answer is yes */
  - relative(Nancy, Matthew)
    /* no answer in general, no if under closed world assumption */
  - $(\exists z)$ ancestor(z, Fred) $\wedge$ ancestor(z, Liz)

# Semantics of FOL

- **Domain M:** the set of all objects in the world (of interest)
- **Interpretation I:** includes
  - Assign each constant to an object in M
  - Define each function of n arguments as a mapping $M^n$ => M
  - Define each predicate of n arguments as a mapping $M^n$ => {T, F}
  - Therefore, every ground predicate with any instantiation will have a truth value
  - In general there is an infinite number of interpretations because |M| is infinite
- **Define logical connectives:** ~, ^, ˇ, =>, <=> as in PL
- **Define semantics of (∀x) and (∃x)**
  - (∀x) P(x) is true iff P(x) is true under all interpretations
  - (∃x) P(x) is true iff P(x) is true under some interpretation

- **Model:** an interpretation of a set of sentences such that every sentence is *True*
- **A sentence is**
  - **satisfiable** if it is true under some interpretation
  - **valid** if it is true under all possible interpretations
  - **inconsistent** if there does not exist any interpretation under which the sentence is true
- **Logical consequence:** S |= X if all models of S are also models of X

# Axioms, definitions and theorems

- **Axioms** are facts and rules that attempt to capture all of the (important) facts and concepts about a domain; axioms can be used to prove **theorems**
  - Mathematicians don't want any unnecessary (dependent) axioms –ones that can be derived from other axioms
  - Dependent axioms can make reasoning faster, however
  - Choosing a good set of axioms for a domain is a kind of design problem
- A **definition** of a predicate is of the form "p(X) $\leftrightarrow$ …" and can be decomposed into two parts
  - **Necessary** description: "p(x) $\rightarrow$ …"
  - **Sufficient** description "p(x) $\leftarrow$ …"
  - Some concepts don't have complete definitions (e.g., person(x))

# More on definitions

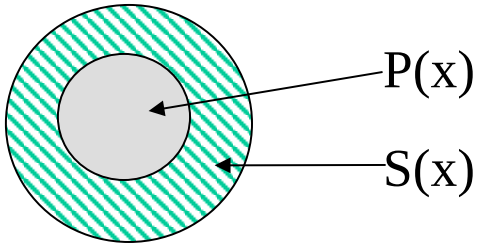
- A **necessary** condition must be satisfied for a statement to be true.
- A **sufficient** condition, if satisfied, assures the statement's truth.
- Duality: "P is sufficient for Q" is the same as "Q is necessary for P."
- Examples: define father(x, y) by parent(x, y) and male(x)
  - parent(x, y) is a necessary (**but not sufficient**) description of father(x, y)
    - father(x, y) → parent(x, y)
  - parent(x, y) ^ male(x) ^ age(x, 35) is a **sufficient** (**but not necessary**) description of father(x, y):

    father(x, y) ← parent(x, y) ^ male(x) ^ age(x, 35)
  - parent(x, y) ^ male(x) is a **necessary and sufficient** description of father(x, y)

    parent(x, y) ^ male(x) ↔ father(x, y)

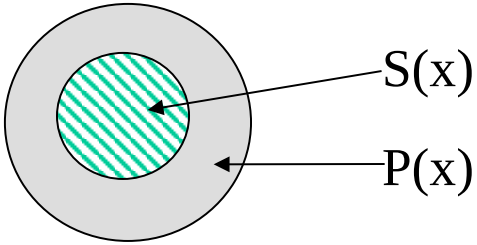
# More on definitions

S(x) is a necessary condition of P(x)



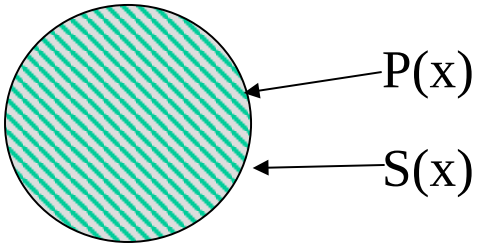

$(\forall x)\ P(x) \Rightarrow S(x)$

S(x) is a sufficient condition of P(x)




$(\forall x)\ P(x) \Leftarrow S(x)$

S(x) is a necessary and sufficient condition of P(x)




$(\forall x)\ P(x) \Leftrightarrow S(x)$

# Higher-order logic

- FOL only allows to quantify over variables, and variables can only range over objects.
- HOL allows us to quantify over relations
- Example: (quantify over functions)

  "two functions are equal iff they produce the same value for all arguments"

  $\forall f \ \forall g \ (f = g) \leftrightarrow (\forall x \ f(x) = g(x))$
- Example: (quantify over predicates)

  $\forall r \ \text{transitive}(r) \rightarrow (\forall xyz) \ r(x,y) \wedge r(y,z) \rightarrow r(x,z))$
- More expressive, but **undecidable**. (there isn't an effective algorithm to decide whether all sentences are valid)
  - First-order logic is decidable only when it uses predicates with only one argument.

# Expressing uniqueness

- Sometimes we want to say that there is a single, unique object that satisfies a certain condition
- "There exists a unique x such that king(x) is true"
  - $\exists x\ king(x) \wedge \forall y\ (king(y) \rightarrow x=y)$
  - $\exists x\ king(x) \wedge \neg\exists y\ (king(y) \wedge x \neq y)$
  - $\exists!\ x\ king(x)$
- "Every country has exactly one ruler"
  - $\forall c\ country(c) \rightarrow \exists!\ r\ ruler(c,r)$
- Iota operator: "$\iota\ x\ P(x)$" means "the unique x such that p(x) is true"
  - "The unique ruler of Freedonia is dead"
  - $dead(\iota\ x\ ruler(freedonia,x))$

# Notational differences

- **Different symbols** for *and*, *or*, *not*, *implies*, ...
  - ∀ ∃ ⇒ ⇔ ^ ˅ ¬ • ⊃
  - p v (q ^ r)
  - p + (q * r)
  - etc
- **Prolog**

  cat(X) :- furry(X), meows (X), has(X, claws)
- **Lispy notations**

```
(forall ?x (implies (and (furry ?x)
                         (meows ?x)
                         (has ?x claws))
                    (cat ?x)))
```

# Logical agents for the Wumpus World

Three (non-exclusive) agent architectures:

- Reflex agents
  - Have rules that classify situations, specifying how to react to each possible situation
- Model-based agents
  - Construct an internal model of their world
- Goal-based agents
  - Form goals and try to achieve them

# A simple reflex agent

- Rules to map percepts into observations:

  $\forall$b,g,u,c,t Percept([Stench, b, g, u, c], t) $\rightarrow$ Stench(t)

  $\forall$s,g,u,c,t Percept([s, Breeze, g, u, c], t) $\rightarrow$ Breeze(t)

  $\forall$s,b,u,c,t Percept([s, b, Glitter, u, c], t) $\rightarrow$ AtGold(t)

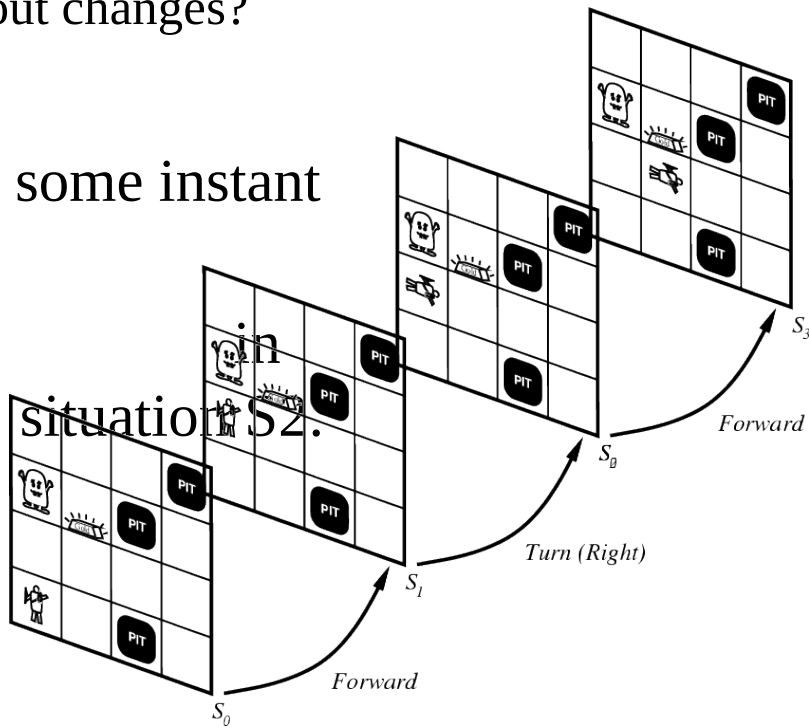
- Rules to select an action given observations:

  $\forall$t AtGold(t) $\rightarrow$ Action(Grab, t);

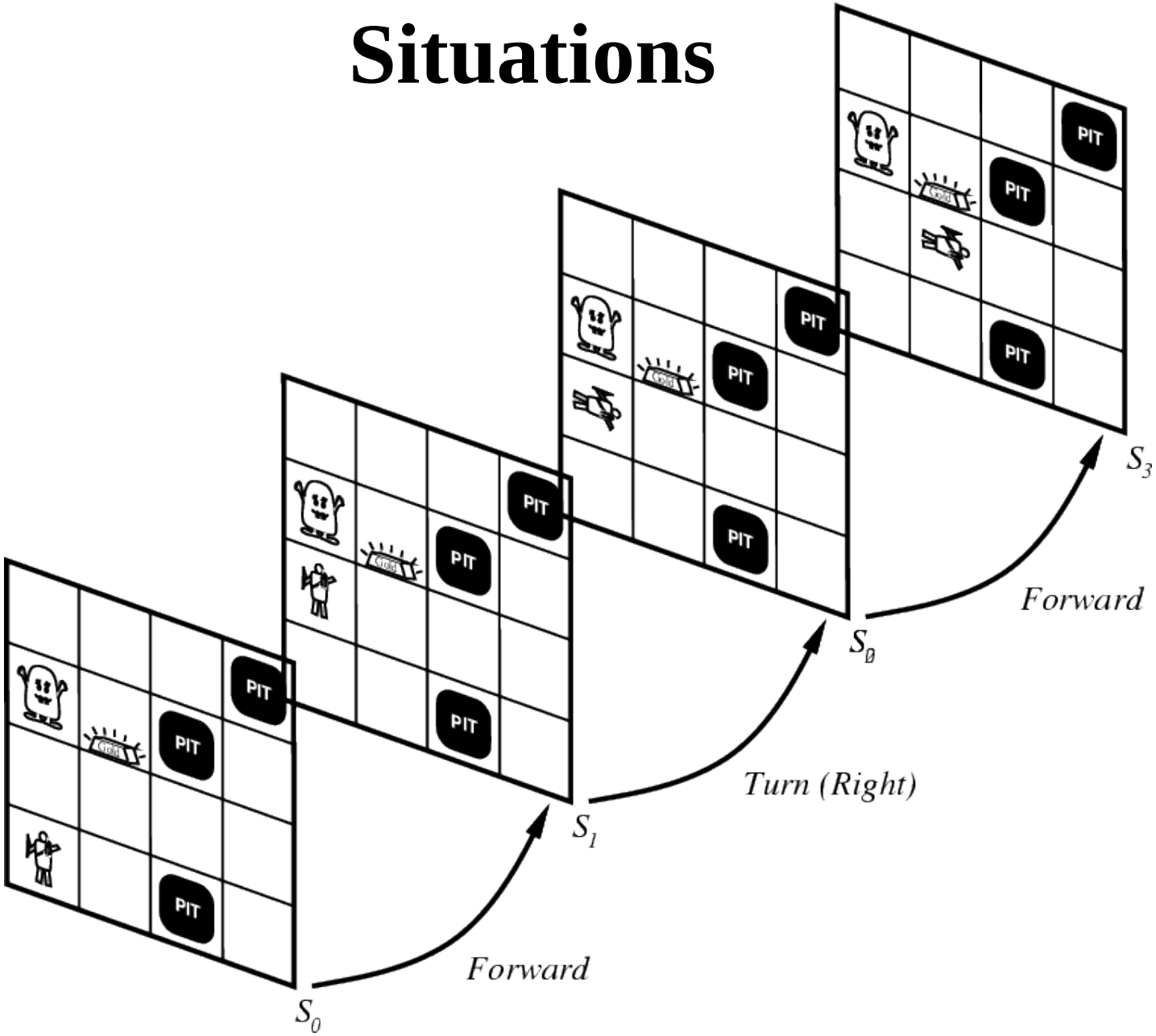
- Some difficulties:
  - Consider Climb. There is no percept that indicates the agent should climb out – position and holding gold are not part of the percept sequence
  - Loops – the percept will be repeated when you return to a square, which should cause the same response (unless we maintain some internal model of the world)

# Representing change

- Representing change in the world in logic can be tricky.
- One way is just to change the KB
  - Add and delete sentences from the KB to reflect changes
  - How do we remember the past, or reason about changes?
- **Situation calculus** is another way
- A **situation** is a snapshot of the world at some instant in time
- When the agent performs an action A in situation S1, the result is a new situation S2.



# Situations


PIT
PIT
PIT
PIT
PIT
PIT
PIT
PIT
PIT
PIT
PIT
PIT
$S_0$
Forward
$S_1$
Turn (Right)
$S_0$
Forward
$S_3$

# Situation calculus

- A **situation** is a snapshot of the world at an interval of time during which nothing changes
- Every true or false statement is made with respect to a particular situation.
  - Add **situation variables** to every predicate.
  - at(Agent,1,1) becomes at(Agent,1,1,s0): at(Agent,1,1) is true in situation (i.e., state) s0.
  - Alternatively, add a special 2nd-order predicate, **holds(f,s),** that means "f is true in situation s." E.g., holds(at(Agent,1,1),s0)
- Add a new function, **result(a,s),** that maps a situation s into a new situation as a result of performing action a. For example, result(forward, s) is a function that returns the successor state (situation) to s
- Example: The action agent-walks-to-location-y could be represented by
  - $(\forall x)(\forall y)(\forall s)\ (at(Agent,x,s) \wedge \neg onbox(s)) \rightarrow at(Agent,y,result(walk(y),s))$

# Deducing hidden properties

- From the perceptual information we obtain in situations, we can **infer properties of locations**

  $\forall l,s\ at(Agent,l,s) \wedge Breeze(s) \rightarrow Breezy(l)$

  $\forall l,s\ at(Agent,l,s) \wedge Stench(s) \rightarrow Smelly(l)$

- Neither Breezy nor Smelly need situation arguments because pits and Wumpuses do not move around

# Deducing hidden properties II

- We need to write some rules that relate various aspects of a single world state (as opposed to across states)
- There are two main kinds of such rules:
  - **Causal rules** reflect the assumed direction of causality in the world:

    $(\forall l1,l2,s)\ At(Wumpus,l1,s) \wedge Adjacent(l1,l2) \rightarrow Smelly(l2)$

    $(\forall\ l1,l2,s)\ At(Pit,l1,s) \wedge Adjacent(l1,l2) \rightarrow Breezy(l2)$

    Systems that reason with causal rules are called **model-based reasoning systems**
  - **Diagnostic rules** infer the presence of **hidden properties** directly from the percept-derived information. We have already seen two diagnostic rules:

    $(\forall\ l,s)\ At(Agent,l,s) \wedge Breeze(s) \rightarrow Breezy(l)$

    $(\forall\ l,s)\ At(Agent,l,s) \wedge Stench(s) \rightarrow Smelly(l)$

# Representing change: The frame problem

- **Frame axioms**: If property x doesn't change as a result of applying action a in state s, then it stays the same.
  - On (x, z, s) ^ Clear (x, s) →
    On (x, table, Result(Move(x, table), s)) ^
    ¬On(x, z, Result (Move (x, table), s))
  - On (y, z, s) ^ y≠ x → On (y, z, Result (Move (x, table), s))
  - The proliferation of frame axioms becomes very cumbersome in complex domains

# The frame problem II

- **Successor-state axiom**: General statement that characterizes every way in which a particular predicate can become true:
  - Either it can be **made true**, or it can **already be true and not be changed**:
  - On (x, table, Result(a,s)) $\leftrightarrow$
    [On (x, z, s) $\wedge$ Clear (x, s) $\wedge$ a = Move(x, table)] $\wedge$
    [On (x, table, s) $\wedge$ a $\neq$ Move (x, z)]
- In complex worlds, where you want to reason about longer chains of action, even these types of axioms are too cumbersome
  - Planning systems use special-purpose inference methods to reason about the expected state of the world at any point in time during a multi-step plan

# Qualification problem

- Qualification problem:
  - How can you possibly characterize every single effect of an action, or every single exception that might occur?
  - When I put my bread into the toaster, and push the button, it will become toasted after two minutes, unless…
    - The toaster is broken, or…
    - The power is out, or…
    - I blow a fuse, or…
    - A neutron bomb explodes nearby and fries all electrical components, or…
    - A meteor strikes the earth, and the world we know it ceases to exist, or…

# Ramification problem

- Similarly, it's just about impossible to characterize every side effect of every action, at every possible level of detail:
  - When I put my bread into the toaster, and push the button, the bread will become toasted after two minutes, and…
    - The crumbs that fall off the bread onto the bottom of the toaster over tray will also become toasted, and…
    - Some of the aforementioned crumbs will become burnt, and…
    - The outside molecules of the bread will become "toasted," and…
    - The inside molecules of the bread will remain more "breadlike," and…
    - The toasting process will release a small amount of humidity into the air because of evaporation, and…
    - The heating elements will become a tiny fraction more likely to burn out the next time I use the toaster, and…
    - The electricity meter in the house will move up slightly, and…

# Knowledge engineering!

- Modeling the “right” conditions and the “right” effects at the “right” level of abstraction is very difficult
- Knowledge engineering (creating and maintaining knowledge bases for intelligent reasoning) is an entire field of investigation
- Many researchers hope that automated knowledge acquisition and machine learning tools can fill the gap:
  - Our intelligent systems should be able to **learn** about the conditions and effects, just like we do!
  - Our intelligent systems should be able to learn when to pay attention to, or reason about, certain aspects of processes, depending on the context!

# Preferences among actions

- A problem with the Wumpus world knowledge base that we have built so far is that it is difficult to decide which action is best among a number of possibilities.
- For example, to decide between a forward and a grab, axioms describing when it is OK to move to a square would have to mention glitter.
- This is not modular!
- We can solve this problem by **separating facts about actions from facts about goals**. This way our **agent can be reprogrammed just by asking it to achieve different goals**.

# Preferences among actions

- The first step is to describe the desirability of actions independent of each other.
- In doing this we will use a simple scale: actions can be Great, Good, Medium, Risky, or Deadly.
- Obviously, the agent should always do the best action it can find:

$(\forall a,s)\ Great(a,s) \rightarrow Action(a,s)$

$(\forall a,s)\ Good(a,s) \wedge \neg(\exists b)\ Great(b,s) \rightarrow Action(a,s)$

$(\forall a,s)\ Medium(a,s) \wedge (\neg(\exists b)\ Great(b,s) \vee Good(b,s)) \rightarrow Action(a,s)$

...

# Preferences among actions

- We use this action quality scale in the following way.
- Until it finds the gold, the basic strategy for our agent is:
  - Great actions include picking up the gold when found and climbing out of the cave with the gold.
  - Good actions include moving to a square that’s OK and hasn't been visited yet.
  - Medium actions include moving to a square that is OK and has already been visited.
  - Risky actions include moving to a square that is not known to be deadly or OK.
  - Deadly actions are moving into a square that is known to have a pit or a Wumpus.

# Goal-based agents

- Once the gold is found, it is necessary to change strategies. So now we need a new set of action values.
- We could encode this as a rule:
  - $(\forall s)$ Holding(Gold,s) $\rightarrow$ GoalLocation([1,1]),s)
- We must now decide how the agent will work out a sequence of actions to accomplish the goal.
- Three possible approaches are:
  - **Inference**: good versus wasteful solutions
  - **Search**: make a problem with operators and set of states
  - **Planning**: to be discussed later